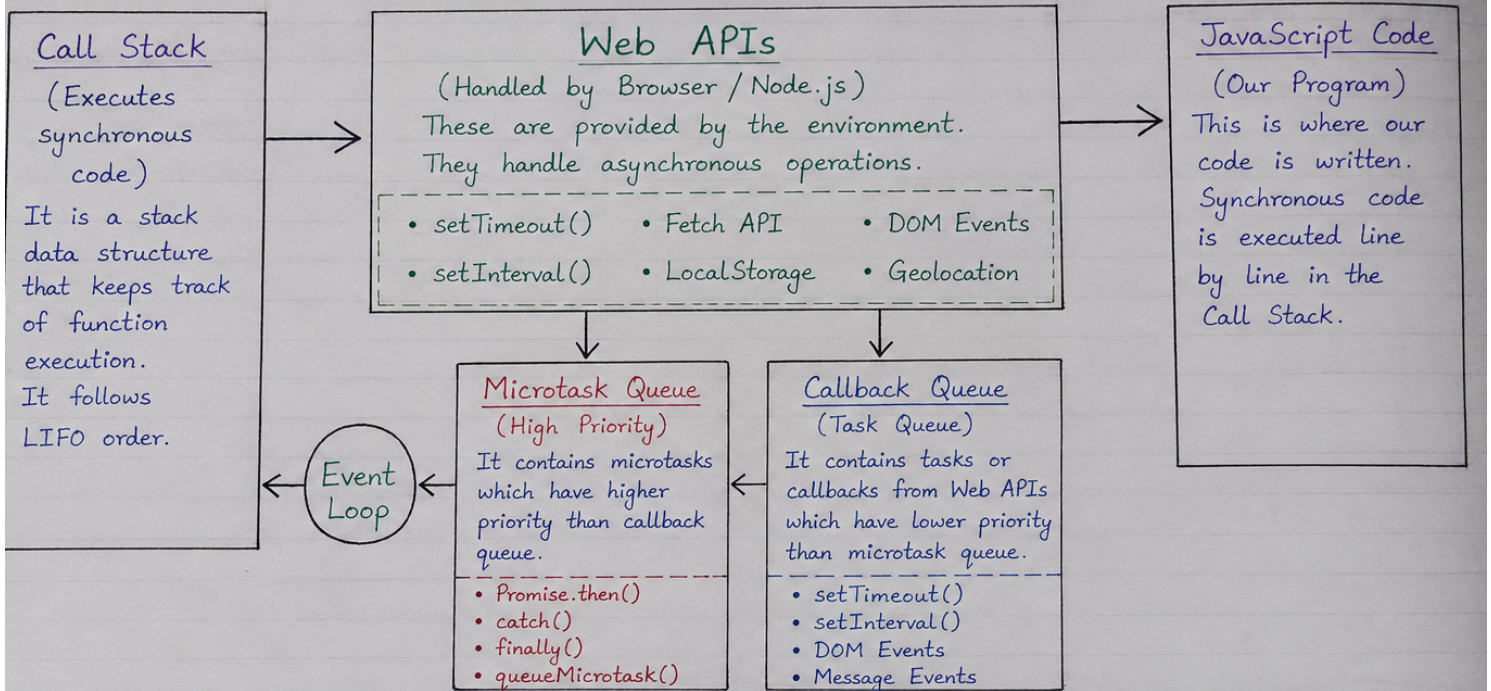


The Event Loop in JavaScript



How Event Loop Works?

JavaScript code is executed in the Call Stack.

When an async operation occurs, it is sent to Web APIs.

Once the operation is completed, its callback is pushed to the appropriate queue:

- Promise callbacks → Microtask Queue
- Other callbacks → Callback Queue

Event Loop continuously checks if Call Stack is empty.

If empty, it first executes all tasks in the Microtask Queue.

After Microtask Queue is empty, it takes one task from Callback Queue and pushes it to Call Stack.

This process repeats continuously.

Priority Order

Call Stack
(Highest)
↓
Microtask Queue
↓
Callback Queue
(Lowest)

Key Points

- JavaScript is single threaded.
- Call Stack can do one thing at a time.
- Microtasks have higher priority.
- Event Loop ensures non-blocking behavior.